DERIVED PUSHFORWARD OF A PERFECT TENSOR A PROPERLY SUPPORTED FLAT COMPLEX IS PERFECT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5804-5817, immediately following the statement at 5793-5802.

Editorial note. The source proof writes 'flat over f^{-1}O_Y' where the statement says flat over B; the flatness input is the same hypothesis.

Allowed standard prerequisites (editorial).

- Perfectness criterion for derived pushforward. Lemma lemma-perfect-direct-image of this chapter, cited at proof lines 5804-5805. Terminal input: Rf_*K is perfect once K is locally bounded with coherent cohomology and finite tor dimension over B.

Source reference. Original lemma 08IS: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5746–5746. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let B be a Noetherian algebraic space over S. Let $f : X \to B$ be a morphism of algebraic spaces which is locally of finite type and quasi-separated. Let $E \in D(\mathcal{O}_X)$ be perfect. Let $\mathcal{G}^\bullet$ be a bounded complex of coherent $\mathcal{O}_X$-modules flat over B with support proper over B. Then $K = Rf_*(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{G}^\bullet)$ is a perfect object of $D(\mathcal{O}_B)$.*

Proof. The object K is perfect by Lemma 08IS. We check the lemma applies: Locally E is isomorphic to a finite complex of finite free $\mathcal{O}_X$-modules. Hence locally $E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{G}^\bullet$ is isomorphic to a finite complex whose terms are of the form

$$\bigoplus\nolimits_{i=a,\ldots,b} (\mathcal{G}^i)^{\oplus r_i}$$

for some integers $a, b, r_a, \ldots, r_b$. This immediately implies the cohomology sheaves $H^i(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{G})$ are coherent. The hypothesis on the tor dimension also follows as $\mathcal{G}^i$ is flat over $f^{-1}\mathcal{O}_Y$. □

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.